Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/13/2005
PI Name:	Lipa, John A. Ph.D.		
Project Title:	Fundamental Physics Experiments With Superconducting Cavity - Stabilized Oscillators on Space Station		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Gravitational & relativistic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94305	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-03
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No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG 3 2852		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
Task Description:	We are developing a new generation of microwave cavities that are much less sensitive to acceleration than earlier models. This should allow us to reach stability levels in the 10^-17 range if other conditions are met. These conditions are improved temperature control and a lower noise phase-locked loop for the cavity. We plan to set up new low temperature systems to meet these requirements. The aim of this work is to develop a superconducting microwave oscillator for conducting relativity experiments on Space Station and to supply a low-noise slave oscillator for the PARCS atomic clock experiment. This task will allow the development of highly stable oscillators that can be used for testing Special and General Relativity, and can be utilized and in other applications.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	
Task Progress:	During the present funding period we continued to refine our understanding of the technical needs of the project to meet our science goals. Low temperature testing of the apparatus disclosed a number of instabilities in the beat note of two cavities. A set of tests was performed and it was concluded that most of the visible problems were likely to be due to the room temperature electronics. Additional thermal controllers were built for temperature-sensitive components and input signals were boosted as much as possible. High frequency transformers were introduced to break a number of ground loops in critical areas. A start was also made on improved power line isolation for the equipment racks. These upgrades led to a marked improvement on the stability of the signal on time scales of up to a few hours. We now have routine performance in the mid-10^-15 range for this time scale which is relevant to a flight experiment. Since the Q's of the cavities used were ~ 2x10^9, this compares well with our earlier measurements in the mid-10^-16 range where we used Q's of ~ 2x10^10. Since our present electronics is lower noise and better designed than the previous version, we expect further gains to be made by continuing our electronics test program. In principle the system is capable of reaching the mid-10^-16 range with the existing cavity Q's, if all sources of interference can be eliminated. One problem area that was identified was a limited tuning bandwidth of the quartz oscillators used in the slave oscillator frequency stabilization circuit. Improved devices with higher bandwidth were ordered, but a six-month delay developed in the delivery. The devices have just been received and testing of their characteristics will begin shortly. Cavity development was continued with much of the effort being put into bringing up a center-mount design cavity made from very high quality (near single crystal) niobium stock. It was hoped that this approach would give us a new way to reach Q's exceeding 10^10 without having to develop a UHV furnace. This is the 5th cavity fabricated in our development program. Good progress was made on this task. The cavity has now been completely fabricated and is awaiting its first etch. Unfortunately the etch facility at SLAC was temporarily shut down in November due to an accident elsewhere on the site and has been unable to reopen. We expect this to be resolved in the next couple of months. The facility has unique expertise in hydrofluoric acid etching at 0°C and has done a significant amount of work for us in the past. A new antenna design was also fabricated for this cavity, giving us the capability to make antennas completely in-house. After etching and initial testing the cavity will be annealed at 1800°C in a commercial furnace and retested. Work was started on analyzing the Lorentz violation signals that might be obtained from a free-flier version of the experiment. This was undertaken because of the apparent total loss of the ISS as a platform for the experiment, with the advent of the Moon and Mars initiative. The Kostelecky and Mewes (2002) formalism was encoded in a symbolic language, Maple, and tests performed to validate the output. We are now experimenting with different rotational transformations to simulate the case of a slowly rotating spacecraft with various orbital alignments. This work is expected to continue throughout most of the coming year. We have also started to look at the possibility of performing an LPI experiment by using a quartz oscillator and a mercury ion clock on the Shuttle. The hope is that such a payload could be made small enough to be installed in a locker and still produce an interesting science result. Some development of the performance of the quartz oscillator would greatly improve the experiment. In summary, we have described the progress with a project to place a pair of superconducting cavity oscillators in orbit on the ISS, or possibly on a free-flier. The cavity oscillator technology has a demonstrated a frequency stability in the mid-10^-16 range for time intervals of 100 to 3000 seconds with high-Q cavities at 1.3 K, competitive with modern atomic clocks. We are now approaching this performance with lower-Q cavities than used originally, easing the fabrication task. A design insensitive to axial acceleration has been demonstrated. On the ISS the cavities can be used alone to test the general Standard Model Extension and in conjunction with atomic clocks to perform tests of relativity and operate as low phase-noise flywheels for atomic fountain experiments. Our analysis of the limitations of superconducting cavity oscillators indicates that improvements to the 10^-17 range are quite possible, given sufficient resources to develop practical fabrication technologies. Analytical work is also in progress to look at the signals obtainable with other configurations of the experiment.
Bibliography Type:	Description: (Last Updated: 11/02/2007)
Articles in Peer-reviewed Journals	C. Lammerzahl, G. Ahlers, N. Ashby, M. Barmatz, P.L. Biermann, H. Dittus, V. Dohm, R. Duncan, K. Gibble, J. Lipa, N. Lockerbie, N. Mulders, C.. Salomon "Experiments in Fundamental Physics Scheduled and in Development for the ISS" General Rel. and Grav., 36, 615 (2004) , May-2004